

HR Attrition Intelligent Recommender

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Problem Statement

- 16% of employees leave every year
- Replacing one employee costs up to 200% of their salary
- HR teams find out too late — after the decision is made



Business Context



\$14.1 M estimated annual attrition cost for this workforce



- 1,470 employees · 16% attrition · ~235 departures/year
- Replacement cost: 1× annual salary (conservative)

A 15% reduction in attrition saves over \$2M annually

Dataset Overview

Employees

Features

Attrition Rate

Missing

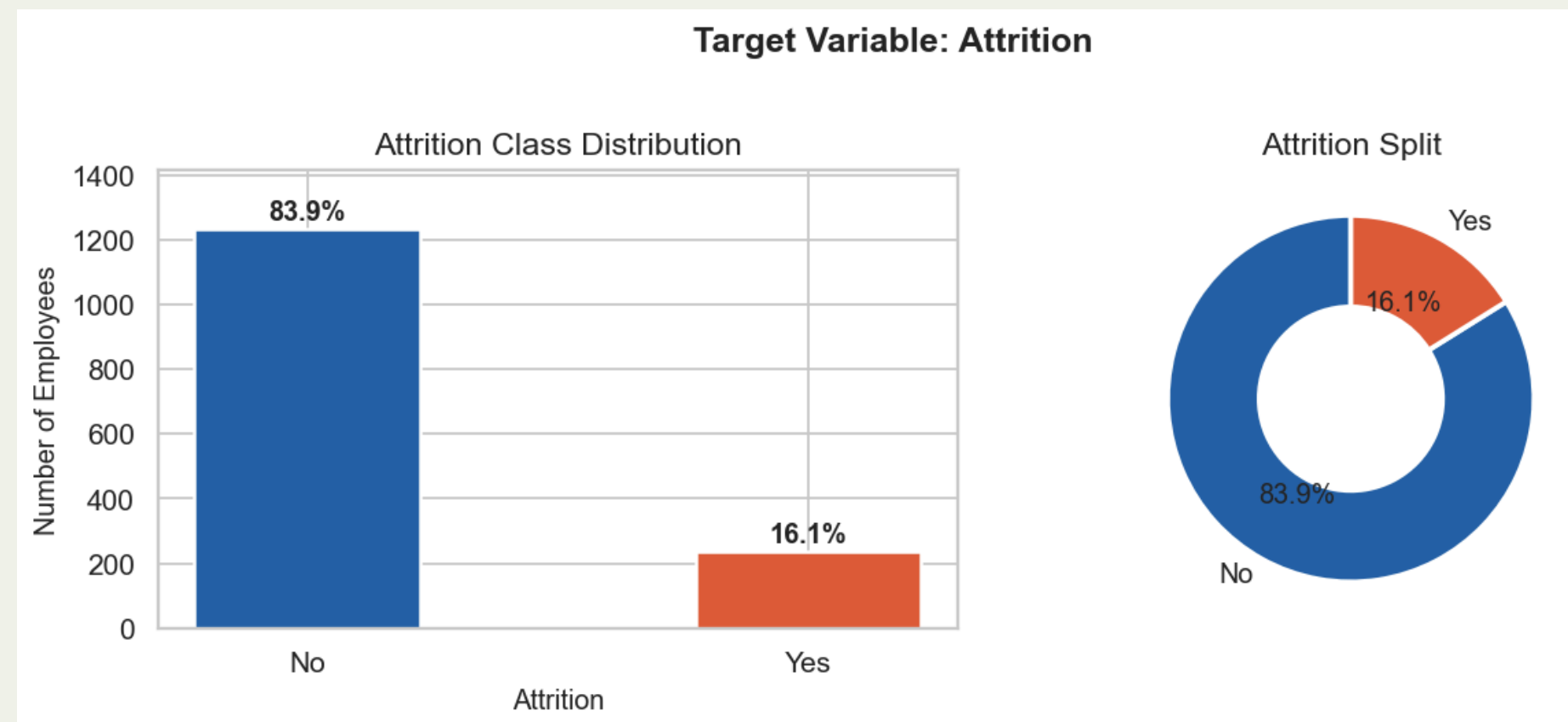
1,470

35

16.10%

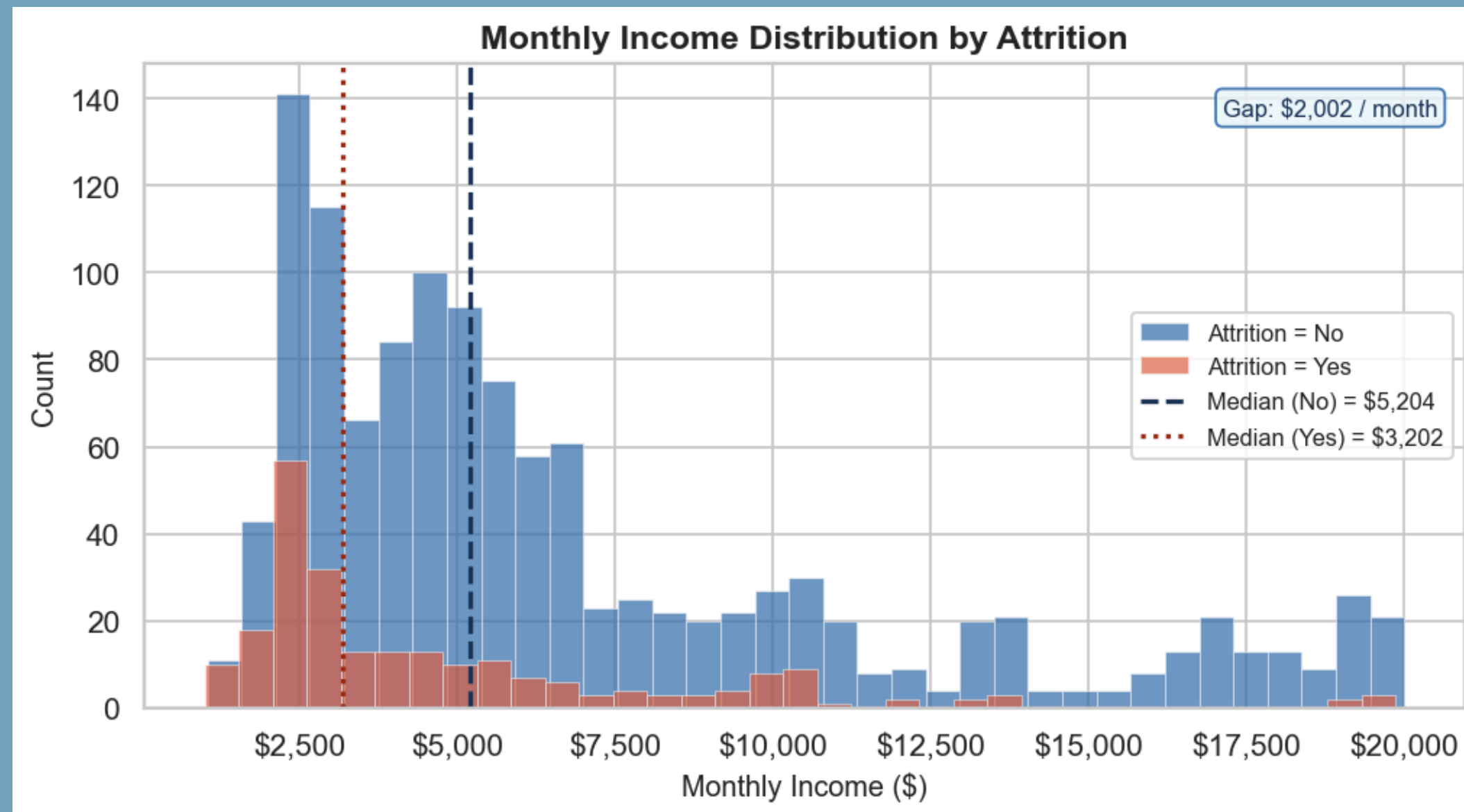
0

35 features covering demographics, compensation, satisfaction, job history



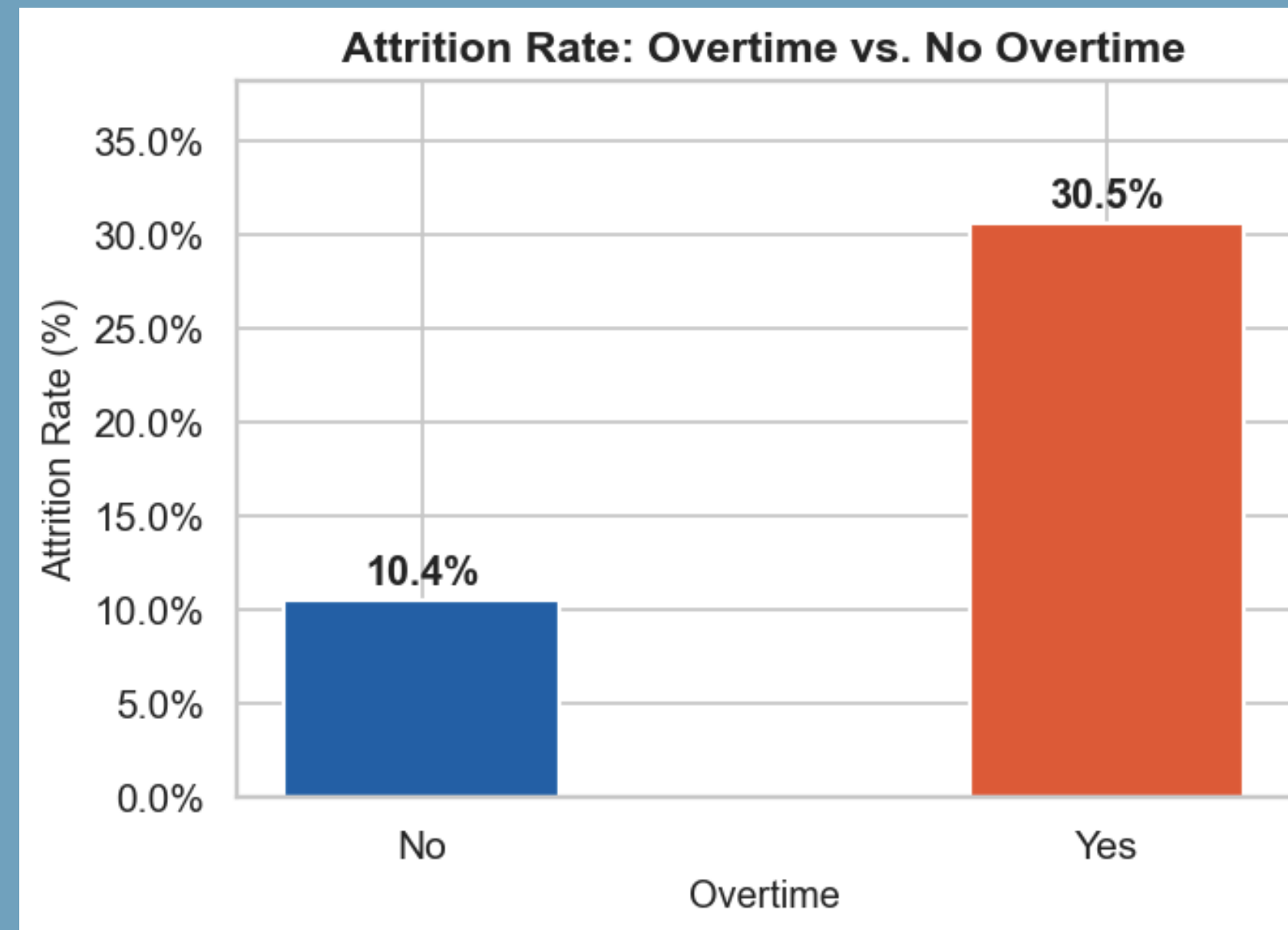
Key EDA Findings I

Pay Gap \$2,002/month Leavers earn significantly less than stayers — same role, different pay



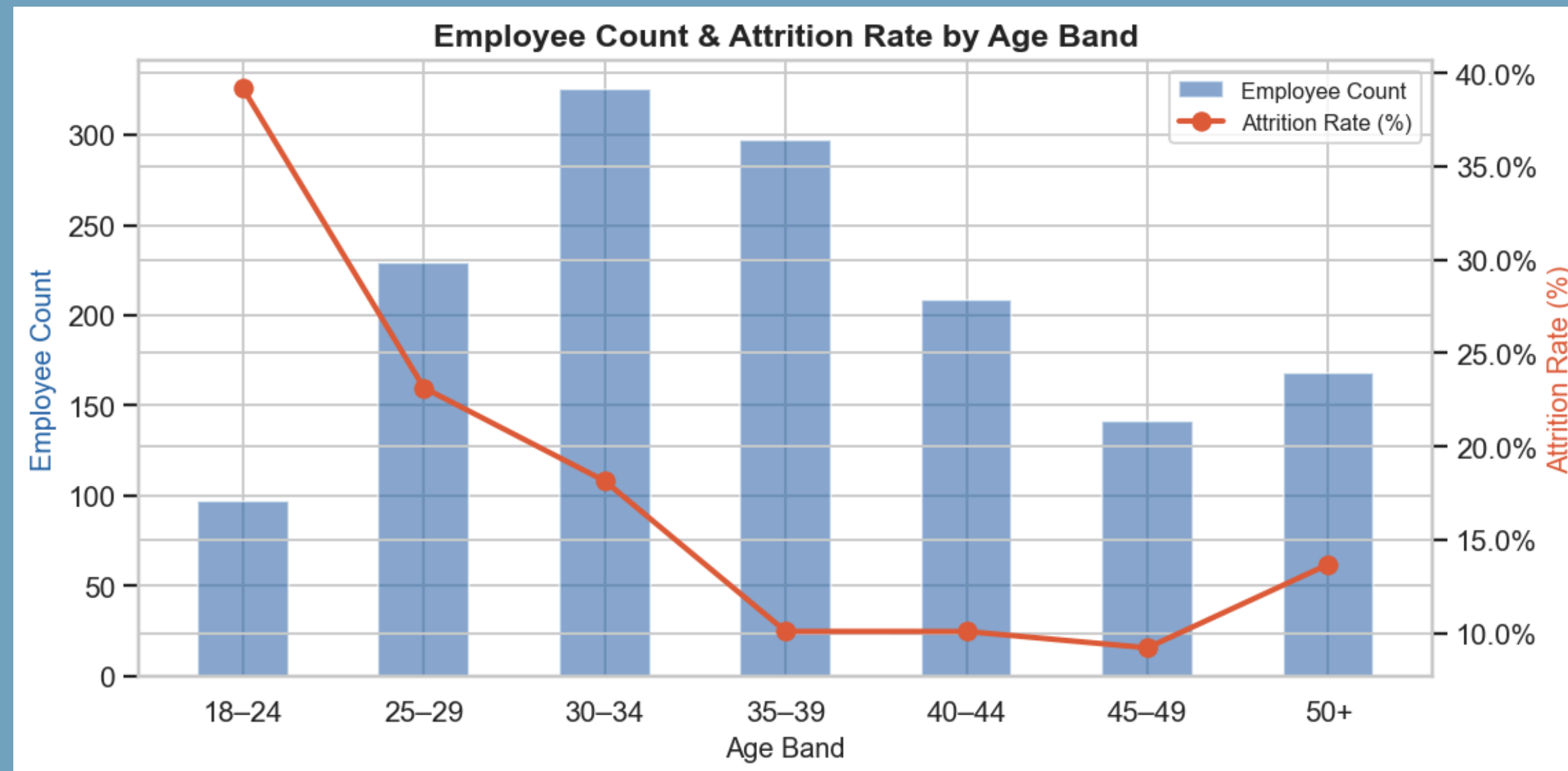
Key EDA Findings II

Overtime 2.9× higher risk Employees working overtime leave at 30.5% vs 10.4%



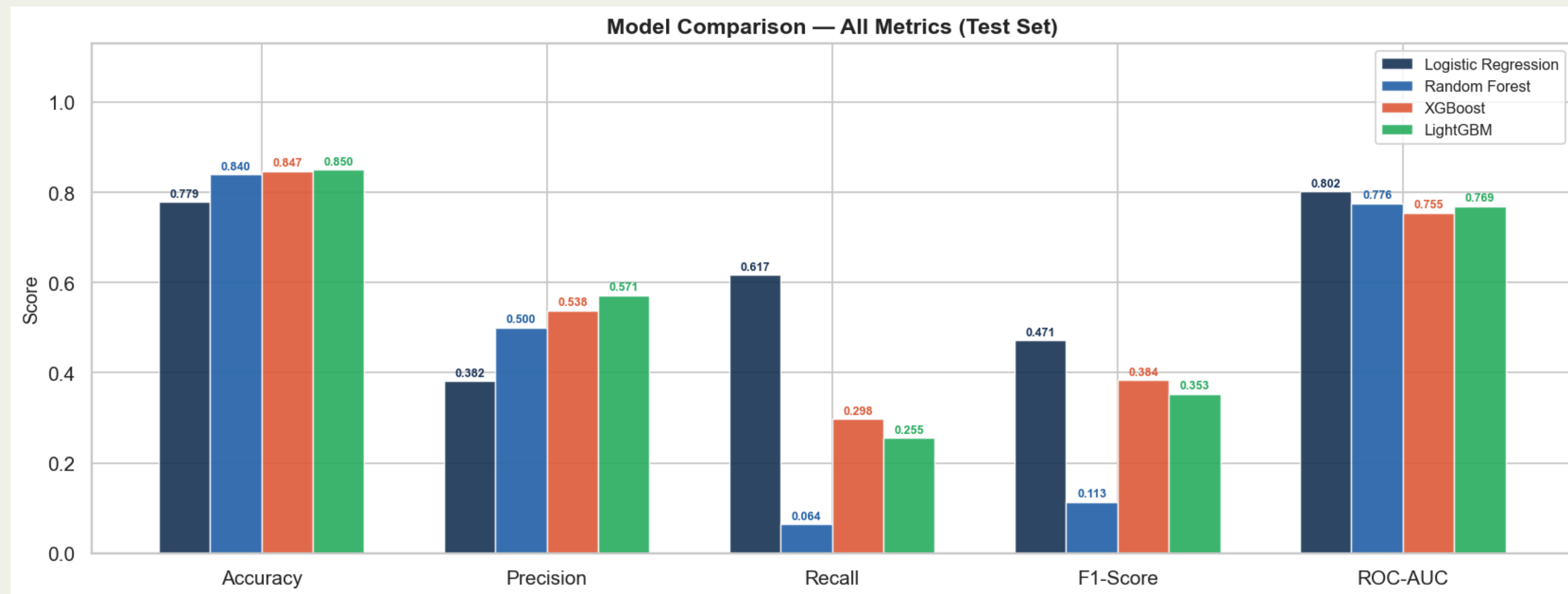
Key EDA Findings III

Age 27.9% vs 10.9% Under-30 employees leave at nearly 3× the rate of over-40s



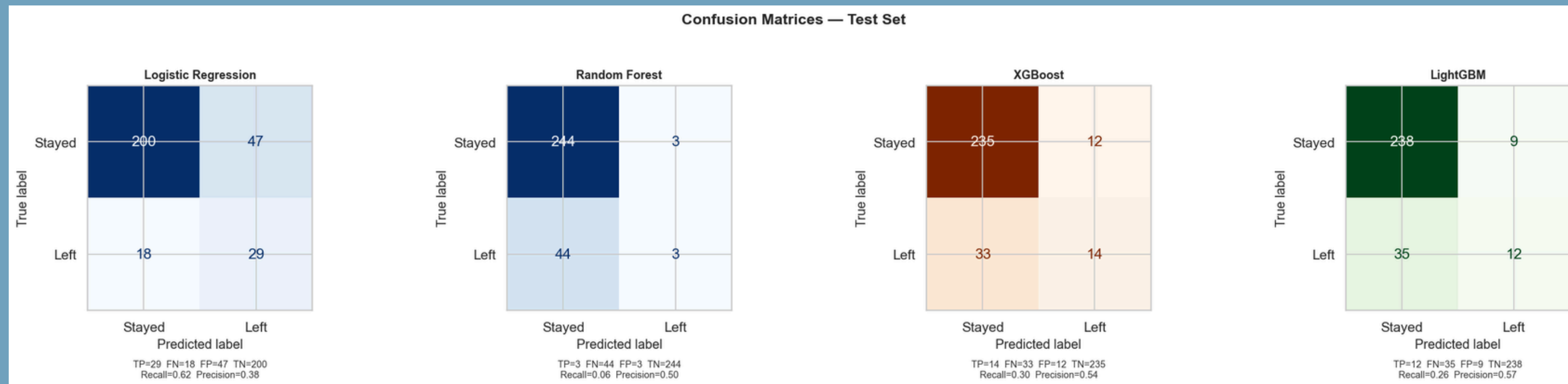
Model Results

Tree models hit 85% accuracy by predicting "Stayed" for almost everyone – they missed 94% of actual leavers



Model Comparison I

Why Accuracy Is Misleading ?



Model Comparison II

Recall = % of actual leavers we catch. Missing a leaver costs 1 year's salary.

Model

Recall

AUC

Logistic Regression

0.681

0.802

Random Forest

0.064

0.776

XGBoost

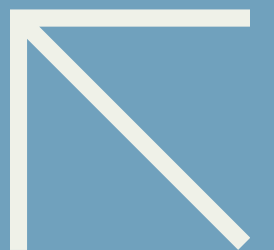
0.298

0.755

LightGBM

0.255

0.769



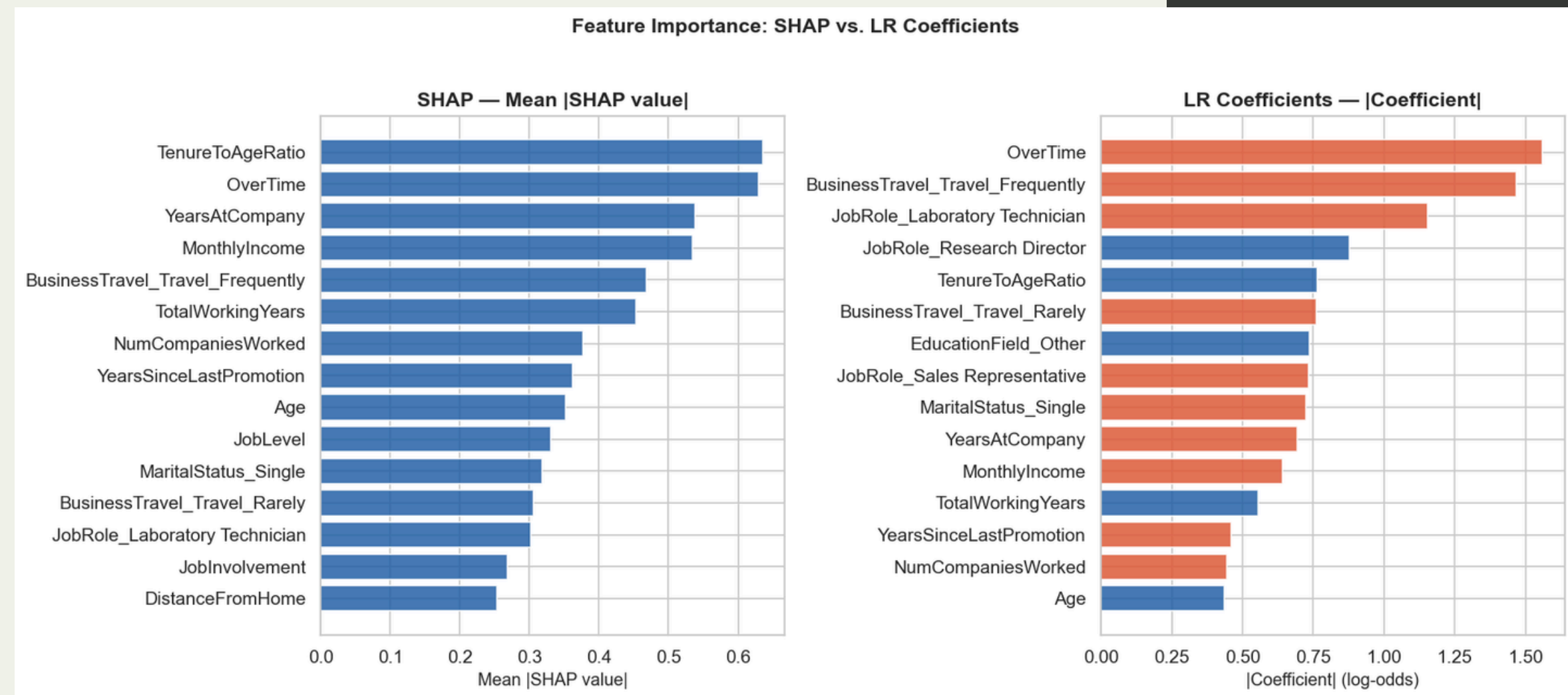
SHAP Global Insights

SHAP values reveal what actually matters — not just what the model learned

#1 TenureToAgeRatio — engineered feature

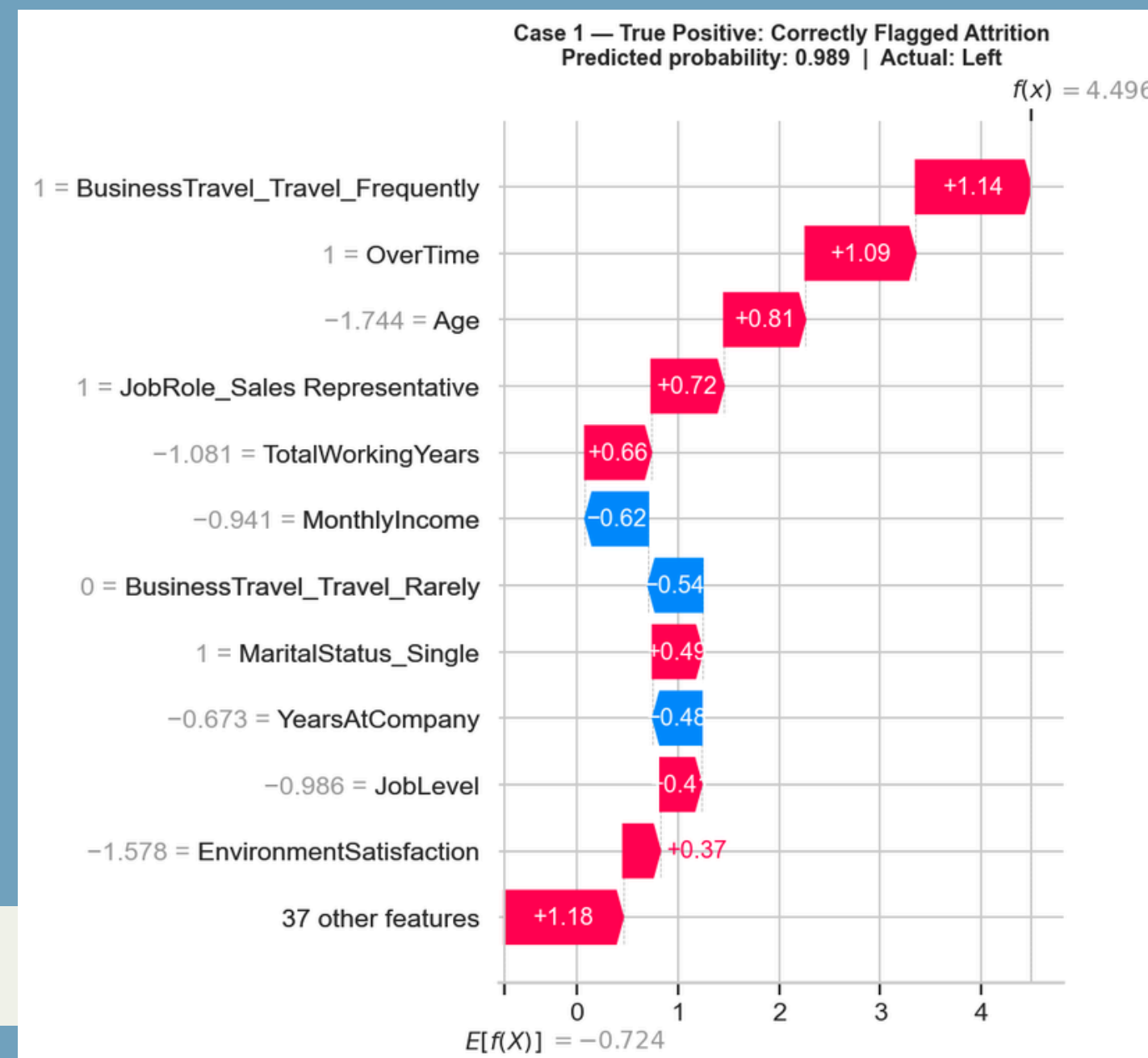
#2 OverTime — largest single positive risk factor

#3 YearsAtCompany — captures stagnation, not loyalty



SHAP Individual Case

Caught (prob 98.9%)



Business Recommendations



**Overtime
reduction policy**

OverTime SHAP #2

High Priority



**Compensation
gap review**

Income gap \$2,002

High Priority



**Satisfaction
pulse survey
(score < 2)**

SatisfactionIndex

Medium Priority



**Early-career
mentorship**

TenureToAgeRatio #1

High Priority

Estimated Impact



\$360K – \$480K Overtime policy (6–8 fewer departures/year)



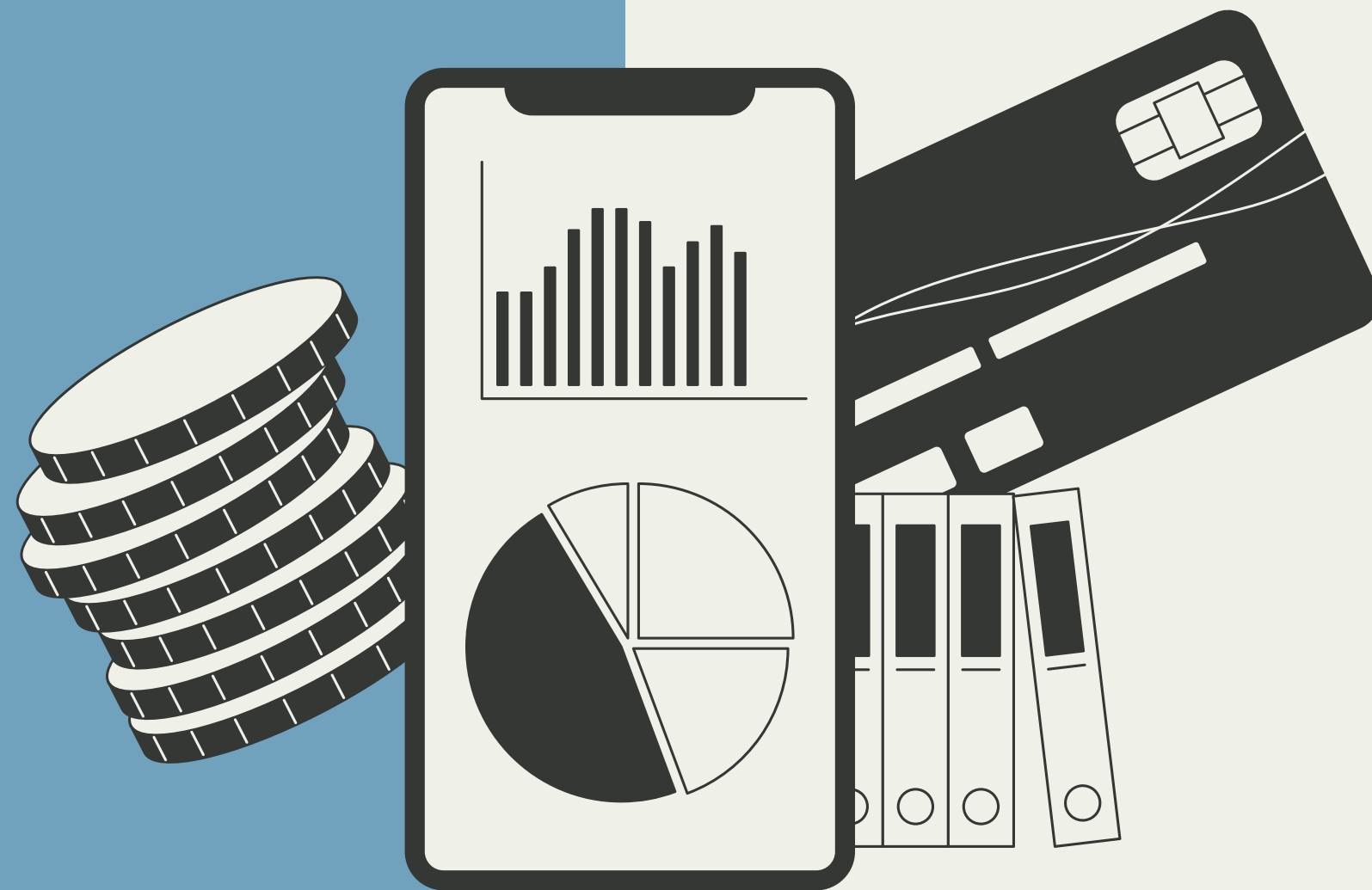
\$900K – \$1.2M Early-career programme (15–20 fewer departures/year)



\$480K – \$600K Satisfaction survey (8–10 fewer departures/year)

All figures use \$60K average salary × 1× replacement cost – conservative estimate

Live Demo



Ethics & Limitations



What We Need to Be Careful About

Risks

MaritalStatus and Gender in the model — demographic bias possible

Managers seeing scores could reduce opportunities for flagged employees

Dataset is synthetic — needs retraining on real data annually

Mitigations

Fairness audits across demographic groups before deployment

Model output goes to HR only — never line managers

SHAP explanations make every decision auditable



Conclusion & Next Steps

3 key takeaways

LR outperforms tree models because feature engineering linearised the key relationships

TenureToAgeRatio (engineered) is the #1 predictor — more powerful than any raw feature

The system correctly flags 68% of employees who will leave — before they decide

next steps

Retrain on live HR data (not synthetic)

Remove demographic features after fairness audit

Integrate with HRIS for monthly automatic scoring

Pilot with one department before full rollout

Thank You